

Your assignment is to design an analytical dashboard for the sales and customer success team to help them understand customer purchasing behavior and order activity. They review the dashboard monthly to monitor customer engagement, identify high-value customers, track order patterns, and detect inactive customers in order to improve customer retention and increase overall sales performance.

The goal is to increase customer retention, improve customer engagement, and maximize sales revenue by understanding customer purchasing behavior and identifying inactive or high-value customers.

# Customer Purchasing Behavior & Retention Analysis

## Business Overview

To better understand customer purchasing behavior and support retention strategies, customers were segmented based on both their purchasing activity and revenue contribution. Customer status was determined using the number of days since their last purchase, classifying customers as Recent, Active, At Risk, or Inactive. Additionally, customers whose total revenue contribution ranked within the top 20% were categorized as High Value customers, while the remaining customers were classified as Low Value. Combining these two dimensions created four strategic customer segments: High Value Active, High Value Inactive, Low Value Active, and Low Value Inactive.

## Key Findings

The business generated approximately \$139 million in revenue from 25,000 orders placed by 5,000 customers, with an average order value of \$5.54K. While these figures indicate strong sales performance, a deeper analysis of customer engagement reveals significant retention challenges.

Purchase frequency analysis shows that most customers make between 4 and 6 purchases, after which customer activity declines sharply. This suggests that while customers are successfully acquired and converted, maintaining long-term engagement remains a challenge.

Customer status analysis further highlights this issue. The largest customer group is Inactive customers (2,091), followed by At Risk customers (1,133). In comparison, only 1,046 customers are currently Active, while 687 customers are classified as Recent customers. This indicates that a substantial portion of the customer base is no longer actively engaging with the business.

The customer value segmentation provides additional insight into revenue performance. Although High Value Active customers contribute significantly to revenue, there are more High Value Inactive customers (552) than High Value Active customers (448). This finding suggests that many of the company's most valuable customers have stopped purchasing, creating a major opportunity for targeted reactivation campaigns.

Revenue contribution by segment reinforces this observation. Low Value Inactive customers generated the highest cumulative revenue (\$55.3M), followed by Low Value Active customers (\$31.2M) and High Value Inactive customers (\$28.6M). The fact that inactive customers account for a large share of historical revenue suggests that customer attrition is affecting segments that once contributed significantly to business growth.

Monthly revenue trends also show that inactive customer segments consistently outperform active segments in cumulative revenue contribution. This indicates that while the business has been successful in generating revenue, many previously valuable customers are no longer actively engaged.

## Business Implications

The analysis suggests that revenue growth is currently being supported by customers who have become inactive over time. This presents a significant retention risk, as the business may increasingly rely on acquiring new customers rather than maximizing the lifetime value of existing ones.

To improve long-term performance, the sales and customer success teams should prioritize reactivation efforts for High Value Inactive and At Risk customers, strengthen loyalty initiatives that encourage repeat purchases, and develop proactive engagement strategies to prevent customer churn. By focusing on retention and customer re-engagement, the business can increase customer lifetime value, improve customer satisfaction, and unlock additional revenue opportunities from its existing customer base.

## Conclusion

The dashboard reveals that while overall sales performance is strong, customer retention remains a critical area for improvement. The combination of a large inactive customer population and a significant number of inactive high-value customers highlights the need for targeted retention and reactivation strategies. By leveraging the customer segmentation framework and focusing on customer engagement, the organization can improve retention rates, strengthen customer relationships, and drive sustainable revenue growth.

| Customer Status | Criteria                                  |
|-----------------|-------------------------------------------|
| Recent          | Last purchase made within 0–30 days       |
| Active          | Last purchase made within 31–90 days      |
| At Risk         | Last purchase made within 91–180 days     |
| Inactive        | Last purchase made more than 180 days ago |
| No Purchase     | No purchase history available             |

| Customer Value | Criteria                                                                           |
|----------------|------------------------------------------------------------------------------------|
| High Value     | Customers whose total revenue falls within the top 20% (80th percentile and above) |
| Low Value      | Customers whose total revenue falls below the 80th percentile                      |

| Customer Segment    | Description                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------|
| High Value Active   | High-revenue customers who are actively purchasing and engaging with the business.                  |
| High Value Inactive | High-revenue customers who have stopped purchasing and represent a key reactivation opportunity.    |
| Low Value Active    | Lower-revenue customers who remain engaged and have potential to increase their spending over time. |
| Low Value Inactive  | Lower-revenue customers with low engagement and limited recent purchasing activity.                 |

| Customer Segment    | Criteria                                                                   |
|---------------------|----------------------------------------------------------------------------|
| High Value Active   | Top 20% revenue customers AND Active (31–90 days since last purchase)      |
| High Value Inactive | Top 20% revenue customers AND Inactive (>180 days since last purchase)     |
| Low Value Active    | Bottom 80% revenue customers AND Active (31–90 days since last purchase)   |
| Low Value Inactive  | Bottom 80% revenue customers AND all remaining statuses not captured above |

